

Cybersecurity Incident Report

Network Traffic Analysis

Part 1: Summary of the problem found in the network traffic log

The Wireshark TCP/HTTP log shows the web server at 192.0.2.1 being flooded with a very large number of TCP SYN requests from a single IP address, 203.0.113.0, all directed at port 443. The attacker's SYN packets arrive repeatedly and rapidly, and the server is unable to keep up. Legitimate employee visitors from the 198.51.100.0/24 range begin receiving connection errors, and from log entry 125 onward the server stops responding to legitimate traffic entirely. Because the flood originates from one source, this is a direct denial of service (DoS) attack in the form of a SYN flood.

Part 2: Analysis of the data and likely cause of the incident

The incident was identified after an automated monitoring alert reported a problem with the web server, and attempts to load the company website returned a connection timeout error. Using a packet sniffer, the security analyst captured the traffic to and from the web server and reviewed the color-coded log, where the attacker's interactions with IP address 203.0.113.0 are marked in red and legitimate visitor traffic is marked in green.

Normal connections complete the TCP three-way handshake (SYN, SYN-ACK, ACK) and then request the sales.html page over HTTP, as seen in the early green entries. The attack traffic breaks this pattern. Starting at log entry 52, the source 203.0.113.0 sends a continuous stream of SYN packets to port 443 without completing the handshake as expected, at a rate of several requests per second. As the volume grows, legitimate visitors start to fail: the server returns RST, ACK packets that drop their connection attempts, and a gateway returns HTTP/1.1 504 Gateway Time-out messages when the server takes too long to respond. By log entry 125, the only traffic the server logs is from the attacker, showing that its resources have been exhausted.

The likely cause is a SYN flood attack, a network-level denial of service in which the attacker sends more SYN requests than the server has resources to handle, consuming the connection capacity reserved for the handshake and preventing legitimate employees from reaching the website. Because all of the malicious traffic comes from the single IP address 203.0.113.0, this is a direct DoS attack rather than a distributed one. Immediate containment was to take the server offline to recover and to block the offending IP address at the firewall, while noting that IP blocking alone is a temporary measure since an attacker can spoof IP addresses to bypass it.